LiteLKG-Net: A Lightweight Network With Multiscale Large Kernel Attention and Gated Spatial Feature Fusion for Remote Sensing Change Detection

Zhihua Hua, Longfei Zhang, Yijian Cai, and Kui Wen

***Abstract*—With the advancement of Earth observation technology, remote sensing change detection (RSCD) has become an effective means to monitor dynamic changes in surface objects, widely applied in urban planning, environmental disaster monitoring, and other fields. Although Transformer-based methods exhibit strong global modeling capabilities, they often suffer from high computational complexity and excessive parameter counts, while traditional convolutional neural network (CNN)-based methods are constrained by limited convolutional kernel sizes, leading to the neglect of global contextual information. These limitations hinder their applicability in resource-constrained and complex scenarios. To address these challenges, this article proposes a LiteLKG-Net, which integrates multiscale large kernel attention (MLKA) and a gating mechanism to maintain high detection performance while significantly reducing computational costs. First, we design a multiscale attention module (MAM) composed of an MLKA and a gated spatial attention unit (GSAU). The MLKA aggregates global and local information to capture rich change features at varying granularities, while the GSAU combines the gating mechanism with spatial attention to fully exploit contextual spatial information with reduced computational overhead. Second, we construct a channel interaction module (CIM) to enhance cross-temporal feature interaction, thereby improving spatiotemporal semantic perception. A feature fusion module (FFM) is further introduced to aggregate multiscale features, enhancing the model's capability to discern fine-grained details. The decoder incorporates a multilevel residual fusion strategy and embeds a gated mechanism module (GMM) to adaptively filter critical features, significantly enhancing model accuracy and robustness. Finally, we adopt MobileNetV2 as the Siamese encoder to efficiently extract features while minimizing computational complexity. Comprehensive experiments are conducted on three benchmark datasets. Both visual and quantitative results demonstrate that our proposed LiteLKG-Net outperforms ten other state-of-the-art networks. In addition, ablation studies validate the effectiveness of each proposed module.**

***Index Terms*—Channel interaction, feature fusion, lightweight networks, multiscale attention, remote sensing change detection (RSCD).**

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I. Introduction

REMOTE sensing change detection (RSCD) aims to identify dynamic changes in surface features within specific geographic regions by analyzing multitemporal remote sensing images (e.g., satellite or aerial imagery) of the same area. As a pivotal technology, RSCD has been extensively deployed in urban planning [1], disaster detection and assessment [2], [3], [4], environmental monitoring [5], [6], and land resource management [7], [8], serving as an indispensable tool for geospatial analytics. RSCD, as a specialized branch of semantic segmentation, fundamentally differs from object detection in nature [9]. RSCD requires simultaneous feature recognition and temporal change analysis, thereby exhibiting greater complexity. Furthermore, RSCD performs pixelwise temporal classification by evaluating change masks using $F1$ score and IoU, while object detection achieves instance-level spatial recognition through bounding box localization, with its performance measured by the mean mAP metric. With the innovation of remote sensing sensors and imaging technology, the resolution of remote sensing images has been greatly improved, including spatial resolution (SR, e.g., WorldView-4 satellite SR: 0.3 m) and temporal resolution (TR, e.g., high-frequency revisiting satellite constellation). While the improved resolution brings rich texture and spatiotemporal information to remotely sensed images [10], it also makes RSCD more challenging [11]. Uncertainty at the data level (data labeling error, category imbalance, and so on) and model level (pseudochange misdetection, small-scale target omission, change region boundary blurring, and so on) have been the key issues plaguing the accuracy and robustness of RSCD. This work focuses on enhancing model accuracy, robustness, and lightweight design. To address these challenges, researchers have proposed numerous advanced models [12], [13], [14] to achieve more reliable detection results.

Over the past decades, RSCD methodologies have evolved from traditional feature-driven frameworks to data-driven deep learning paradigms. Conventional approaches, predominantly categorized as pixel-oriented and object-oriented methods, rely on manual feature extraction and mathematical modeling. Pixel-based methods, such as image differencing [15], ratio analysis [16], principal component analysis (PCA) [17], and change vector analysis (CVA) [18], generate difference

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maps by directly comparing corresponding pixel features, followed by empirically defined thresholding to derive final change detection results [19]. Despite their computational efficiency and interpretability, these approaches face significant performance constraints due to their dependence on handcrafted feature engineering, heightened sensitivity to noise, and limited generalization capacity across diverse environmental conditions. The object-based approach analyzes changes by segmenting and further fusing multidimensional features, such as spectrum, texture, and shape [20], [21], [22], [23], which has the advantages of suppressing pixel-level noise, enhancing spatial continuity, and supporting semantic-level change interpretation. However, the method relies heavily on the segmentation quality and suffers from the problems of cumbersome adjustment of parameters and fine change miss detection, which limits its application efficiency in large-scale complex scenes. Compared to the above change detection methods, RSCD methods based on traditional machine learning have made some progress, such as support vector machine [24], K-means [25], and random forest regression [26], which improve the detection accuracy and robustness by automatically learning nonlinear features and fusing multisource data. However, they are still limited by the difficulty of advanced feature acquisition, insufficient computational efficiency, and reliance on manual feature design, and have a limited ability to generalize to complex scenes.

The rapid development of remote sensing data and graphics processing units (GPUs) has significantly enhanced the application of deep learning techniques in RSCD. Convolutional neural networks (CNNs), as a deep learning architecture, exhibit powerful end-to-end learning, multilayer nonlinear complex modeling, and flexible scalability [27], [28], [29], making them widely adopted in RSCD. The local receptive fields and parameter-sharing mechanism of CNNs enable automated extraction of localized features while drastically reducing parameter counts, thereby lowering overfitting risks. In addition, through hierarchical feature extraction, CNNs acquire low-level features (edges, colors) and high-level semantic features (object parts and overall shapes), rendering them suitable for complex visual tasks involving high-dimensional data. However, traditional CNNs face inherent limitations due to the restricted local receptive fields of convolutional kernels: weak long-range dependence modeling and insufficient global contextual information interaction. Recently, Transformer networks [30], [31], [32], [33] have achieved remarkable success in RSCD by leveraging self-attention mechanisms to overcome local receptive field constraints, capturing long-range dependencies to model global contextual relationships. While the self-attention mechanism enhances global modeling capabilities, it substantially increases computational resource consumption, leading to limited practicality in real-time monitoring scenarios. Furthermore, Transformer-based change detection models require large-scale training datasets to fully exploit their performance advantages, yet their application is constrained by the high cost of data annotation.

To address challenges posed by data-level uncertainties and limitations of existing models, we propose LiteLKG-Net, a lightweight network for RSCD. First, LiteLKG-Net employs a pretrained MobileNetV2 [34] as its backbone for feature extraction. This architecture innovatively integrates inverted residual structures, demonstrating superior performance in semantic segmentation and object detection tasks while achieving an optimal balance between accuracy and efficiency. Compared to classical networks like ResNet [35] and VGG [36], MobileNet V2 exhibits significant advantages in computational efficiency and parameter reduction. Second, to enhance detection accuracy for fine-grained changes (especially fine variations) and object boundaries, LiteLKG-Net introduces a multiscale attention module (MAM) comprising multiscale large kernel attention (MLKA) and a gated spatial attention unit (GSAU). The MLKA component captures global long-range dependencies and local detail features, significantly improving the model's adaptability to complex scenes. Simultaneously, the GSAU utilizes a gating mechanism and depthwise separable convolution to generate dynamic spatial weights, focusing on critical regions while suppressing noise, thereby balancing feature enhancement and raw information retention. To address false detections caused by nonsemantic changes (e.g., illumination variations and shadows), we introduce the channel interaction module (CIM). This module employs a lightweight periodic channel interaction mechanism to achieve efficient information exchange between bitemporal input features, balancing feature diversity with computational efficiency. In change detection tasks, the CIM significantly enhances the model's sensitivity to genuine change regions while maintaining low computational overhead. Finally, to further improve detection accuracy and robustness, LiteLKG-Net incorporates a feature fusion module (FFM). By leveraging co-occurrence feature enhancement and a lightweight dynamic fusion strategy, this module achieves efficient interaction and difference amplification between bitemporal features, substantially boosting sensitivity to change regions and localization precision with minimal computational cost. The decoder incorporates a multilevel residual fusion strategy and embeds a gated mechanism module (GMM) to adaptively filter critical features, significantly enhancing model accuracy and robustness. The primary contributions of this work are as follows.

1) We propose LiteLKG-Net, a novel lightweight network for RSCD, designed to address critical challenges of data-level uncertainties (e.g., annotation errors and class imbalance) and model-level uncertainties (e.g., false positives, missed small-scale targets, and blurred change boundaries), thereby enhancing detection accuracy and robustness.
2) The MAM innovatively integrates MLKA and GSAU. The MLKA captures long-range dependencies and local detail features via multiscale large-kernel convolutions (3×3, 5×5, and 7×7), hierarchically extracting contextual information to enhance adaptability to complex scenes. The GSAU employs a gating mechanism and depthwise separable convolution to adaptively select spatial features with low computational complexity, significantly improving spatial detail modeling capabilities.

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3) The CIM facilitates cross-temporal feature interaction through periodic channel exchange, enhancing feature diversity and complementarity to better capture local inconsistencies between bitemporal inputs, thereby increasing sensitivity to genuine changes. The FFM strengthens co-occurrence features via elementwise multiplication, residual connections, and cascaded nonlinear activations, suppressing irrelevant background noise while emphasizing true changes. The decoder incorporates a multilevel residual fusion strategy and embeds a GMM to adaptively filter critical features, significantly enhancing model accuracy and robustness.

We conduct comprehensive experiments on three benchmark datasets (LEVIR-CD, WHU-CD, and SYSU-CD) and compare LiteLKG-Net with ten state-of-the-art methods, validating its feasibility and superiority. Ablation studies further confirm the effectiveness of each module in LiteLKG-Net.

II. Related Work

A. CNN-Based RSCD Methods

With the advancement of deep learning, RSCD has gradually shifted from traditional feature engineering to end-to-end deep learning model-driven approaches. As a critical component of deep learning, CNNs are increasingly applied to RSCD [37], [38], [39], [40]. Daudt et al. [41] pioneered the use of dual-branch fully convolutional networks for RSCD, where twin encoders independently extract bitemporal features and fuse them via concatenation (FC-Siam-conc) or difference (FC-Siam-diff). While validating the effectiveness of dual-branch architectures, their method suffers from simplistic feature interaction, weak global modeling, and strict reliance on precise image registration. Subsequent studies have focused on optimizing feature fusion and enhancing contextual semantics. Chen and Shi [42] proposed STANet, which employs a spatiotemporal attention module to dynamically weight change-sensitive regions, improving detection accuracy. Chen et al. [43] developed a dual-branch U-Net with weight-shared encoders for bitemporal feature extraction and fused multilevel difference maps to balance detail preservation and feature decoupling in high-resolution RSCD. Wang et al. [44] utilized a multiscale pyramidal convolutional encoder and cross-domain feature alignment to harmonize structural relationships and semantic features, enabling effective change region identification. Despite these advancements, existing methods remain constrained by weak long-range dependence modeling, insufficient global contextual interaction, and limited robustness to registration errors, hindering their deployment in large-scale complex scenarios.

B. Transformer-Based RSCD Methods

In recent years, Transformer architectures have emerged as a research focus in RSCD due to their global self-attention mechanism. ChangeFormer [45] pioneered the application of pure Transformer architectures to RSCD, capturing spatiotemporal correlations via hierarchical Transformer encoders and multilayer perceptron (MLP) decoders. However, its computational complexity grows quadratically with image resolution, making it impractical for high-resolution remote sensing imagery. To balance efficiency and performance, hybrid architectures have gained prominence. The BIT model [46] employs CNN encoders to extract local features and a Transformer decoder to fuse global context, achieving advanced levels on multiple benchmarks. Nevertheless, its high training resource demands and limited real-time performance remain critical challenges. Concurrently, researchers have explored lightweight designs (e.g., ChangeRD [47], which reduces computational costs via attention-guided central difference convolution) and multimodal fusion (e.g., integrating optical and SAR data to mitigate cloud interference [48]), further advancing practical deployment. Despite these advancements, Transformer-based methods still suffer from insufficient sensitivity to local details, leading to high miss rates for small targets. They also have higher computational complexity and larger parameter counts compared to CNN-based methods, hindering deployment on resource-constrained edge devices.

C. Weakly Supervised and Unsupervised RSCD Methods

Beyond supervised learning, weakly supervised and unsupervised approaches offer novel solutions to reduce label dependence. CDGAN [49] leverages generative adversarial networks (GANs) to synthesize difference maps, minimizing reliance on pixel-level annotations. Sun et al. [50] constructed an unsupervised CD framework using nonlocal patch similarity, effectively reducing false alarms in heterogeneous image CD. Zhao et al. [51] proposed a plug-and-play dense instance separation (DISep) method to enhance detection accuracy in weakly supervised settings. Liu et al. [52] developed an unsupervised change detection framework based on image translation, improving model generalization on unannotated data. However, such methods still exhibit lower accuracy compared to supervised counterparts and demonstrate insufficient stability in complex scenarios. In addition, emerging spiking neural network (SNN) models have shown strong potential for low-power and ultralow-latency visual analysis in remote sensing [53], [54], [55], offering promising directions for future lightweight or neuromorphic CD frameworks.

Despite significant technological advancements, RSCD remains challenged by false positives, missed small-scale targets, blurred change boundaries, and high computational complexity. To address these issues, this article proposes LiteLKG-Net, a lightweight RSCD framework. First, LiteLKG-Net employs a pretrained MobileNetV2 [34] backbone for feature extraction, achieving an optimal balance between performance and efficiency. Second, to enhance detection accuracy for fine-grained changes (particularly tiny variations) and object boundaries, we design a MAM composed of MLKA and a GSAU, which significantly improves adaptability to complex scenes. To mitigate false alarms caused by nonsemantic changes (e.g., illumination variations and shadows), we introduce a CIM that balances feature diversity and computational efficiency. Finally, to further boost detection accuracy and robustness, an FFM is proposed to amplify discriminative features through efficient interaction and difference enhancement between bitemporal inputs. The decoder incorporates a multilevel residual fusion strategy and embeds

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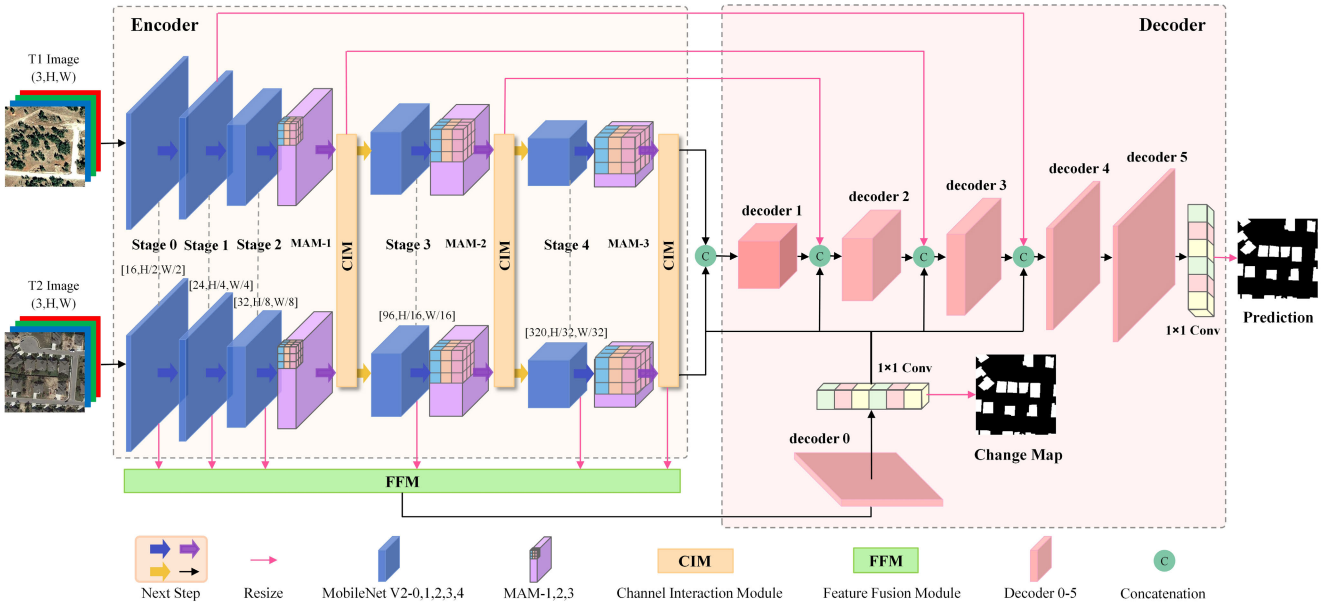


Fig. 1. Schematic of the proposed LiteLKG-Net architecture.

a GMM to adaptively filter critical features, significantly enhancing model accuracy and robustness.

III. Methodology

First, the overall architecture of the proposed LiteLKG-Net model is outlined in Section III-A. Then, key components of the LiteLKG-Net encoder are detailed in Sections III-B–III-D. Finally, Section III-E elaborates on the decoder composition and critical modules.

A. Network Architecture Overview

We propose LiteLKG-Net, a lightweight RSCD network that achieves high-precision detection and lightweight deployment through hierarchical feature modeling and dynamic interaction mechanisms. The overall architecture and inference details are illustrated in Fig. 1 and Algorithm 1.

First, LiteLKG-Net adopts a pretrained MobileNetV2 [34] backbone. Its inverted residual structures significantly reduce parameter counts while extracting hierarchical features (16–320 channels) from bitemporal remote sensing images via multistage downsampling. This provides multiscale features balancing semantic abstraction and detail preservation, achieving an optimal performance-efficiency tradeoff.

Second, to enhance detection accuracy for different fine-grained changes (especially subtle variations) and object boundaries, we design a MAM comprising MLKA and a GSAU. The MLKA captures long-range global dependencies and local detail features, significantly improving adaptability to complex scenes. The GSAU generates dynamic spatial weights through a gating mechanism and depthwise separable convolution, prioritizing critical regions while suppressing noise, thereby balancing feature enhancement and raw information retention.

Algorithm 1 Implementation Process of the Proposed LiteLKG-Net

Input : Bi-temporal RS images $A, B \in \mathbb{R}^{H\times W\times 3}$
Output: Change detection map $M_{\text{final}} \in \mathbb{R}^{H\times W}$

// step 1 : Feature Extraction Stage
for $k = 1$ **to** 4 **do**
  $F_A^{(k)}, F_B^{(k)} \leftarrow \text{MobileNetV2}_{\text{features}[k]}(A),\ \text{MobileNetV2}_{\text{features}[k]}(B)$
  if $k \geq 2$ **then**
    $F_A^{(k)}, F_B^{(k)} \leftarrow \text{MAB}^{(k)}(F_A^{(k)}),\ \text{MAB}^{(k)}(F_B^{(k)})$
    $F_A^{(k)}, F_B^{(k)} \leftarrow \text{CIM}(F_A^{(k)}, F_B^{(k)})$
// step 2 : Feature Fusion & Decoding
$F_{\text{fuse}} \leftarrow \text{FFM}(\text{Concat}(F_A^{(1:4)}),\ \text{Concat}(F_B^{(1:4)}))$
$M_{\text{coarse}} \leftarrow \text{Decoder}_{\text{main}}(F_{\text{fuse}})$
// step 3 : Multi-level Refinement
for $l = 4$ **to** 1 **do**
  $F_{\text{up}} \leftarrow \text{BilinearUpsample}(F_{\text{prev}})$
  $F_{\text{fuse}}^{(l)} \leftarrow \text{Concat}(F_{\text{up}},\ F_A^{(l)},\ F_B^{(l)},\ M_{\text{coarse}})$
  $F_{\text{refined}}^{(l)} \leftarrow \text{SqueezeDoubleConv}(F_{\text{fuse}}^{(l)})$
$M_{\text{final}} \leftarrow \sigma(\text{Decoder}_{\text{final}}(F_{\text{refined}}^{(1)}))$

To address false alarms caused by nonsemantic changes (e.g., illumination variations and shadows), a CIM is introduced. This module periodically exchanges feature channels between dual branches at configurable intervals, enhancing sensitivity to genuine changes through differential information fusion. To further improve accuracy and robustness, the FFM implements efficient bitemporal interaction and discriminative enhancement via co-occurrence feature reinforcement and lightweight dynamic fusion, significantly boosting change

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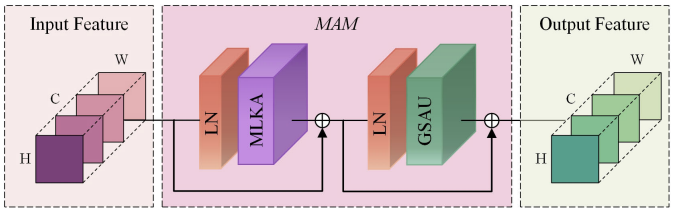


Fig. 2. Overview of the MAM.

sensitivity and localization precision with minimal computational overhead.

Finally, the decoder adopts a multilevel residual fusion strategy to dynamically aggregate high-level semantic features with low-level detail information, and embeds a GMM to adaptively filter key features, significantly improving detection accuracy and robustness in complex scenes while maintaining low computational overhead.

B. MAM

The MAM serves as the core component of LiteLKG-Net, designed to enhance precise localization of change regions in remote sensing imagery through multiscale contextual perception and dynamic spatial attention optimization. As depicted in Fig. 2, the MAM comprises two critical submodules: the MLKA and the GSAU. The MLKA captures long-range global dependencies and local detail features via multibranch dilated convolutions, enhancing the model's adaptability to complex scenes. The GSAU generates dynamic spatial weights using depthwise separable convolution and a gating mechanism, suppressing noise while prioritizing critical regions. By synergizing MLKA and GSAU, the MAM achieves a balance between computational efficiency and feature representation capability within a lightweight framework, significantly improving change detection accuracy and robustness.

Given input features X, the overall processing flow of the MAM is as follows:

$$\begin{aligned} L &= \mathrm{LN}(X) \\ X &= X + \lambda_1 f_3\left(\mathrm{MLKA}\left(f_1(L)\right) \otimes f_2(L)\right) \\ L &= \mathrm{LN}(X) \\ X &= X + \lambda_2 f_6\left(\mathrm{GSAU}\left(f_4(L), f_5(L)\right)\right) \end{aligned} \tag{1}$$

where $LN(\bullet)$ denotes layer normalization, λ is a learnable scaling factor, MLKA($\bullet$) and GSAU($\bullet$) represent the proposed **MLKA** and **GSAU** modules, respectively, $\otimes$ indicates elementwise multiplication, and $f_i(\bullet)$ denotes the ith pointwise convolution layer for preserving feature dimensions.

1) Multiscale LKA: Attention mechanisms dynamically focus on critical information while improving parallel computational efficiency. Previous RSCD models have incorporated various attention mechanisms (e.g., channel attention [56], spatial attention [57], and self-attention [58]) to enhance feature representation. However, these methods typically rely on fixed-receptive-field attention maps, failing to capture local details and long-range dependencies simultaneously. As shown in Fig. 3, the MLKA module aims to efficiently capture global

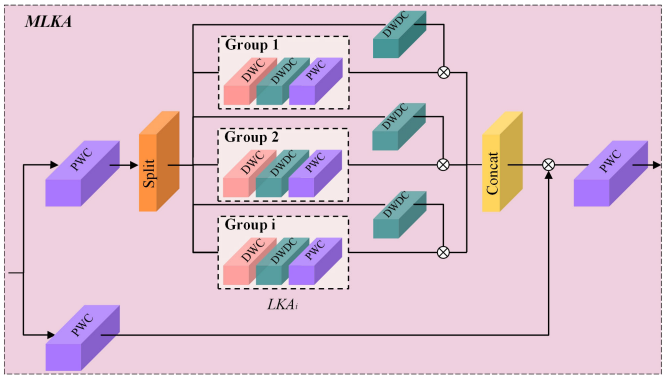


Fig. 3. Architecture of our MLKA.

context and local information in remote sensing imagery through a synergistic design of large-kernel decomposition and multiscale learning, thereby enhancing the model's ability to perceive subtle changes in complex scenes. This module processes input features via a parallel multibranch architecture integrated with a residual learning mechanism, significantly improving feature representation robustness while maintaining lightweight computational overhead. Specifically, the MLKA comprises three components: large kernel attention (LKA) to establish dependence relationships; multiscale partitioning to associate and extract multiscale features, and gated aggregation that dynamically calibrates attention maps to enhance local feature retention.

a) Large Kernel Attention: LKA adaptively constructs long-range dependencies relationship through a convolution kernel decomposition strategy, decomposing traditional large-kernel convolutions (e.g., $K \times K$) into three components: depthwise convolution, depthwise-dilated convolution, and pointwise convolution. Given input features map $X \in \mathbb{R}^{C\times H\times W}$, the specific operations are as follows:

$$\mathrm{LKA}(X) = f_{\mathrm{PW}}\left(f_{\mathrm{DWD}}\left(f_{\mathrm{DW}}(x)\right)\right) \tag{2}$$

where f_{DW} denotes a depthwise convolution with a kernel size of $(2d-1)\times(2d-1)$, f_{DWD} denotes a depthwise-dilated convolution with a kernel size of $\lceil k/d\rceil \times \lceil \frac{k}{d}\rceil$ (with a dilation rate of d), and f_{PW} denotes a pointwise convolution.

b) Multiscale Groupwise Mechanism: To effectively capture multiscale information encompassing both local details and global context, we introduce a multiscale groupwise mechanism based on fixed LKA. Serving as the core for efficient context modeling, this mechanism performs channelwise partitioning and multibranch parallel processing, enabling fine-grained extraction of target features through the combination of convolutional kernels and dilation rates at various scales. First, given an input feature map $X \in \mathbb{R}^{C\times H\times W}$, the module divides it into n pieces $X_1, X_2, \ldots, X_n$ ($X_i \in \mathbb{R}^{n/C\times H\times W}$) along the channel dimension. By partitioning the feature space into independent subspaces, each group focuses on feature extraction at a specific scale, reducing the computational load of a single branch and mitigating interference between features of different scales. For the ith group X_i, an LKA module parameterized by $\{K_i, d_i\}$ is applied to generate a homogeneous-scale attention map LKA_i. Specifically, we adopt three groups of

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LKA settings: {7, 2} implemented by 3–5–1, {21, 3} by 5–7–1, and {35, 4} by 7–9–1, where a–b–1 denotes sequential $a \times a$ depthwise convolution, $b \times b$ depthwise-dilated convolution, and 1×1 pointwise convolution. Finally, the outputs from all branches are concatenated along the channel dimension and integrated with a gated aggregation factor and a residual connection to balance the enhanced features with the original input

$$\mathrm{MLKA}_i(X_i) = G_i(X_i) \otimes \mathrm{LKA}_i(X_i) \tag{3}$$

where $G_i(\cdot)$ denotes the gated aggregation factor; $\mathrm{LKA}_i(\cdot)$ represents the output processed by the a–b–1 decomposed LKA for the ith branch.

Algorithm 2 Multiscale Attentional Module (MAM)

Input: Feature map $X \in \mathbb{R}^{C\times H\times W}$
Output: Refined feature map $Y \in \mathbb{R}^{C\times H\times W}$

// **Stage 1: Multi-scale Large Kernel Attention (MLKA)**
$X_{\mathrm{norm}} \leftarrow \mathrm{LayerNorm}(X)$
$X_{\mathrm{proj}} \leftarrow \mathcal{C}_{1\times1}(X_{\mathrm{norm}})$
$(A, X_{\mathrm{main}}) \leftarrow \mathrm{Split}_C(X_{\mathrm{proj}})$
$(A_1, A_2, A_3) \leftarrow \mathrm{Split}_C(A, 3)$
$Y_1 \leftarrow \mathcal{G}^{3\times3}_{C/3}(A_1) \rightarrow \mathcal{D}^{5\times5}_{d=2} \rightarrow \mathcal{C}_{1\times1}$
$Y_2 \leftarrow \mathcal{G}^{5\times5}_{C/3}(A_2) \rightarrow \mathcal{D}^{7\times7}_{d=3} \rightarrow \mathcal{C}_{1\times1}$
$Y_3 \leftarrow \mathcal{G}^{7\times7}_{C/3}(A_3) \rightarrow \mathcal{D}^{9\times9}_{d=4} \rightarrow \mathcal{C}_{1\times1}$
$A_{\mathrm{attn}} \leftarrow \mathrm{Concat}\left(Y_1 \odot \mathcal{G}^{3\times3}_{C/3}(A_1), Y_2 \odot \mathcal{G}^{5\times5}_{C/3}(A_2), Y_3 \odot \mathcal{G}^{7\times7}_{C/3}(A_3)\right)$
$X_{\mathrm{MLKA}} \leftarrow \left[\mathcal{C}_{1\times1}(X_{\mathrm{main}} \odot A_{\mathrm{attn}})\right] \odot \alpha + X$
// **Stage 2: Gated Spatial Attention Unit (GSAU)**
$X_{\mathrm{norm}} \leftarrow \mathrm{LayerNorm}(X_{\mathrm{MLKA}})$
$X_{\mathrm{ext}} \leftarrow \mathcal{C}_{1\times1}(X_{\mathrm{norm}})$
$(A_{\mathrm{sp}}, X_{\mathrm{main}}) \leftarrow \mathrm{Split}_C(X_{\mathrm{ext}})$
$W_{\mathrm{gate}} \leftarrow \sigma\left(\mathcal{G}^{7\times7}_C\left(\mathcal{G}^{7\times7}_C(A_{\mathrm{sp}})\right)\right)$
$Y \leftarrow \left[\mathcal{C}_{1\times1}(W_{\mathrm{gate}} \odot X_{\mathrm{main}})\right] \odot \beta + X_{\mathrm{MLKA}}$
return Y

2) Gated Spatial Attention Unit: Inspired by [59], [60], [61], and [62], we integrate simple spatial attention (SSA) and gated linear unit (GLU) to construct the GSAU for adaptive gating. The GSAU module dynamically prioritizes critical regions through a lightweight spatial attention mechanism while suppressing noise interference caused by illumination variations and shadows. As shown in Fig. 4, the input feature X is first processed by layer normalization and pointwise convolution to expand the channels, and then split into an attention branch and a main feature branch. The key process of GSAU can be formulated as follows:

$$\mathrm{GSAU}(X) = f_{\mathrm{DW}}(X) \otimes X \tag{4}$$

where $f_{\mathrm{DW}}(X)$ denotes the depthwise convolution, and $\otimes$ represents the elementwise multiplication.

C. Channel Interaction Module

To further enhance the encoder's feature extraction capability, we introduce the CIM. This module implements a lightweight feature-interaction mechanism that periodically exchanges channels between bitemporal feature maps, enhancing feature diversity and model sensitivity to changes. As shown in Fig. 5, given input features $X_1, X_2 \in \mathbb{R}^{C\times H\times W}$, the CIM first generates a mask vector of length C. It then performs channel exchange at predefined periodic intervals

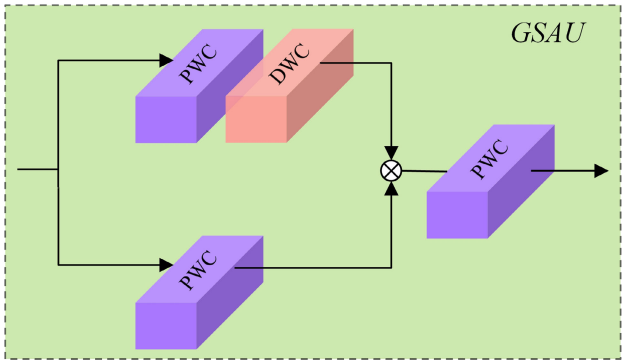


Fig. 4. Architecture of our GSAU.

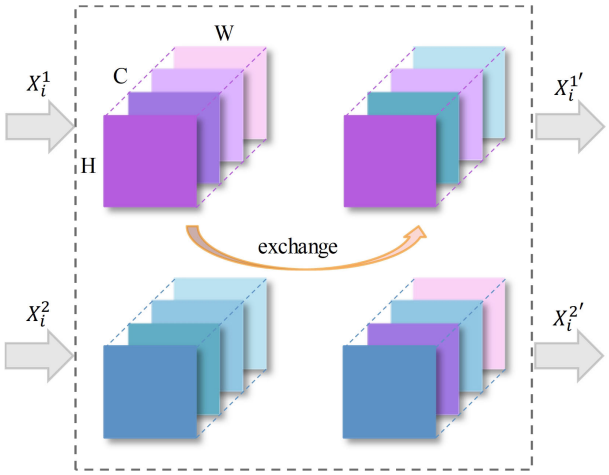


Fig. 5. Architecture of our CIM.

while strictly preserving the original feature distribution of nonselected channels. By interactively fusing feature channels, the CIM effectively breaks the representational consistency of unchanged regions, guides the model to focus on the inherent differences between bitemporal remote sensing images, and significantly enhances the model's robustness against nonreal changes, such as illumination differences and seasonal changes. In addition, the reorganization of information along the channel dimension requires no extra parameters or convolutional operations, thereby improving model accuracy while maintaining a lightweight design.

D. Feature Fusion Module

To fully leverage multistage feature information, an FFM is constructed. As illustrated in Fig. 6, the FFM serves as the core component in LiteLKG-Net for efficient bitemporal feature interaction. Through a cascaded residual fusion strategy, the module enhances the recognition of small-scale changes and improves detection performance while maintaining a lightweight architecture. First, bitemporal multilevel features $F_1^0, \ldots, F_1^4$ and $F_2^0, \ldots, F_2^4$ obtained from the encoder are concatenated to form preliminary feature combinations x_1 and x_2. Second, the concatenated features x_1 and x_2 pass through a shared 1×1 convolutional layer for channel adjustment, aligning dimensionalities and reducing redundant

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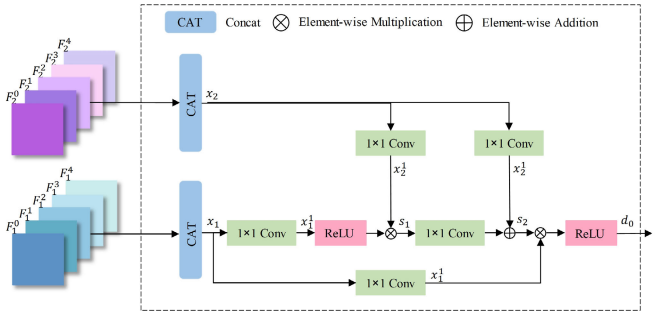


Fig. 6. Architecture of our FFM.

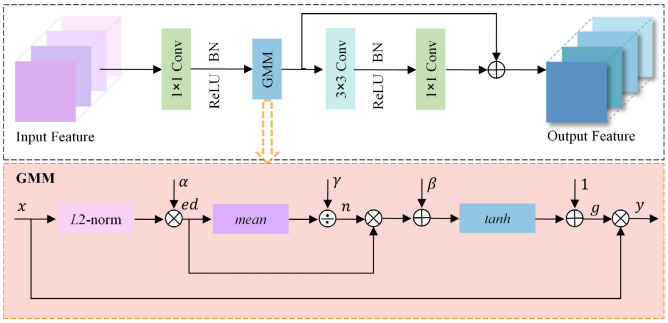


Fig. 7. Architecture of our GMM.

channel information, thereby further decreasing computational complexity. Feature x_1^1 is processed by a ReLU activation function to enhance nonlinear representation and multiplied elementwise with feature x_2^1, thereby improving the sensitivity to small-scale changes. The calculation is as follows:

$$s_1 = \mathrm{ReLU}\left(x_1^1\right) \otimes x_2^1 \tag{5}$$

where s_1 denotes the fused feature in the first stage, and x_1^1 and x_2^1 represent the features obtained by applying a 1×1 convolution operation on the fused features x_1 and x_2, respectively.

By introducing residual connections, feature s_1 is processed through a 1×1 convolution operation and then added to the original feature x_2^1, preserving the original information while establishing a baseline for the difference, thereby enhancing the model's ability to perceive change information. This process can be expressed by the following formula:

$$s_2 = \mathrm{Conv}\left(s_1\right) \oplus x_2^1 \tag{6}$$

where s_2 denotes the fused feature in the second stage.

Finally, the mixed feature s_1 is multiplied elementwise with the feature obtained by applying a 1×1 convolution to the original feature x_1, which significantly enhances change information. The ReLU activation function is subsequently applied for overall refinement, yielding the high-contrast fused feature d_0. This process is formulated as follows:

$$d_0 = \mathrm{ReLU}\left(s_2 \otimes x_1^1\right) \tag{7}$$

where d_0 denotes the final fused feature output by the FFM.

E. Decoder

The decoder of LiteLKG-Net employs a multilevel progressive fusion architecture. As shown in Fig. 7, this architecture enhances change feature identification through hierarchical feature aggregation and a GMM, while maintaining a lightweight design. First, a 1×1 convolution compresses channel dimensions to eliminate redundancies and reduce computational complexity. The GMM then dynamically adjusts channel weights to strengthen feature discrimination and representation capabilities. Subsequently, a 3×3 convolution extracts spatial–semantic features and integrates contextual semantic information. Finally, a 1×1 convolution performs feature decoding, and a residual structure fuses features processed by the GMM to ensure stable gradient propagation.

The GMM serves as the core component in the decoder architecture. It significantly enhances discriminative capability for RSCD through a channel-adaptive feature calibration mechanism. The GMM dynamically adjusts spatial feature weights via a triplet of learnable parameters (α, β, and γ), strengthening responsiveness to change features while maintaining computational efficiency. First, the input features are processed with an $L2$-norm and multiplied by the channel importance scaling factor α, achieving effective aggregation of channel information. The scaling factor α assesses the importance of each channel to efficiently utilize channel information. A small constant term ε ($0 \le \varepsilon \le 10^{-5}$) is introduced during $L2$-norm calculation to preserve computational stability. This process is formulated as follows:

$$ed = \alpha \left\|x_c\right\|_2 = \alpha_c \left\{ \left[\sum_{i=1}^{H} \sum_{j=1}^{W} \left(x_c^{i,j}\right)^2 \right] + \varepsilon \right\}^{\frac{1}{2}} \tag{8}$$

where the input features $X \in \mathbb{R}^{C \times H \times W}$, x_c represents the cth channel of x, and α_c denotes the learnable weight of channel c.

Subsequently, the output feature ed undergoes mean normalization and is divided by the feature norm modulation coefficient γ. The parameter γ is utilized to learn competitive relationships among channels. This process significantly reduces parameter redundancy and is formulated as follows:

$$n = \frac{\gamma_c}{\sqrt{\mathrm{mean}\ (\ ed\ )^2 + \varepsilon}} \tag{9}$$

where γ_c represents the learnable weight of channel c.

Furthermore, the nonlinear bias term β highlights important channel features while suppressing irrelevant ones. A gating function enables nonlinear adjustment of channelwise weights and dynamic feature selection, significantly enhancing the model's capability to discern change boundaries and other complex features. This process is formulated as follows:

$$g_c = 1 + \tanh\left(ed \cdot n + \beta_c\right) \tag{10}$$

where β_c represents the learnable weight of channel c.

Finally, the gating coefficient c is elementwise multiplied with the input features x, yielding the output features y. This process is formulated as follows:

$$y = x \otimes g. \tag{11}$$

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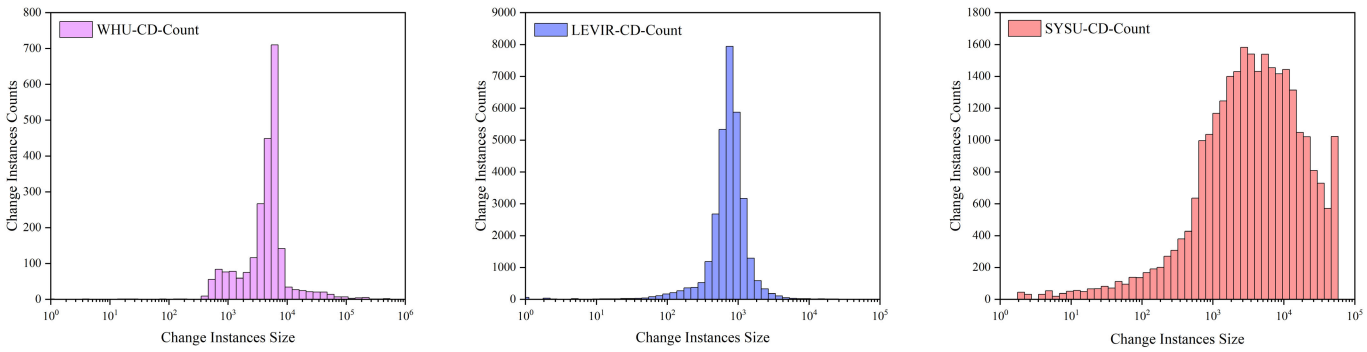


Fig. 8. Distribution of change-instance sizes (pixels).

IV. Experiments

In this section, we evaluate the proposed method on the WHU-CD, LEVIR-CD, and SYSU-CD benchmark datasets to validate its performance and domain adaptability. All experiments were conducted on Ubuntu 18.04.5 LTS using PyTorch 2.4 and Python 3.10, with training performed on an NVIDIA RTX 2080 Ti (22 GB VRAM). We adopt the Adam optimizer with an initial learning rate of 10^{-3} and a cosine annealing schedule. The batch size and number of epochs are set to 32 and 200, respectively, and the model achieving the highest validation $F1$ score is selected for testing.

A. Datasets

To validate the effectiveness and generalizability of the proposed method, three widely adopted public benchmark datasets for change detection were employed in the experiments: WHU-CD [63], LEVIR-CD [42], and SYSU-CD [64].

1) *WHU-CD:* The WHU-CD [63] covers Christchurch, New Zealand, containing one pair of high-resolution aerial images with dimensions of 32 507 × 15 354 pixels and an SR of 0.075 m.
2) *LEVIR-CD:* LEVIR-CD [42] comprises 637 pairs of high-resolution remote sensing images acquired via the Google Earth API. Each image measures 1024 × 1024 pixels with an SR of 0.5 m. The dataset spans 5–14 years, during which significant land-use changes occurred, particularly newly built structures. LEVIR-CD includes diverse building types and large-scale variation in object sizes, effectively testing change detection performance.
3) *SYSU-CD:* SYSU-CD [64] contains 20 000 pairs of aerial images (256 × 256 pixels) captured between 2007 and 2014 in Hong Kong, with a resolution of 0.5 m. Primary change types include: 1) new urban constructions; 2) suburban expansion; 3) preconstruction groundwork; 4) vegetation changes; 5) road extensions; and 6) marine construction.

Due to GPU memory limitations, the high-resolution images in the original datasets cannot be directly used for model training or evaluation. Therefore, we preprocess the data by tiling the remote sensing images into 256 × 256 patches (zero overlap), and then splitting them into training, validation, and testing sets at a ratio of 7:1:2.

TABLE I
Sample Distribution and Class Imbalance Statistics of the Benchmark Datasets

Dataset	Change Instances	Change Pixels	Change Ratio (%)
WHU-CD	2,320	21,354,815	4.28
LEVIR-CD	31,291	31,066,643	4.65
SYSU-CD	27,938	286,092,024	21.83

To assess dataset imbalance, we analyzed the sample distributions of the three benchmark datasets. As shown in Table I, WHU-CD and LEVIR-CD exhibit severe class imbalance, with change ratios of only 4.28% and 4.65%. SYSU-CD has a higher proportion of changed pixels (21.83%) but remains imbalanced. Fig. 8 further shows the distribution of change-instance sizes: WHU-CD and LEVIR-CD are dominated by small to medium instances with concentrated ranges, whereas SYSU-CD presents a much broader, more dispersed distribution with a clear long tail toward large instances. These results highlight substantial differences in scale diversity and imbalance severity across the datasets.

B. Evaluation Metrics

We employ $F1$ score, mean intersection over union (IoU), precision (Pre), recall (Rec), and overall accuracy (OA) as the performance evaluation metrics for the experimental results. Computational complexity (FLOPs), number of parameters (Params), and frames per second (FPS) are utilized as the metrics for evaluating model efficiency. The calculation formulas are as follows:

$$\mathrm{F1} = 2 \times \frac{\mathrm{Pre} \times \mathrm{Rec}}{\mathrm{Pre} + \mathrm{Rec}} \tag{12}$$

$$\mathrm{IOU} = \frac{TP}{TP + FN + FP} \tag{13}$$

$$\mathrm{Pre} = \frac{TP}{TP + FP} \tag{14}$$

$$\mathrm{Rec} = \frac{TP}{TP + FN} \tag{15}$$

$$\mathrm{OA} = \frac{TP + TN}{TP + TN + FN + FP} \tag{16}$$

$$\mathrm{FPS} = \frac{1}{\mathrm{AIT}} \tag{17}$$

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where TP denotes true positives (the number of correctly predicted changed pixels), TN represents true negatives (the number of correctly predicted unchanged pixels), FP signifies false positives (the number of erroneously predicted changed pixels), and FN indicates false negatives (the number of erroneously predicted unchanged pixels). AIT represents the average inference time (the AIT per image).

C. Comparison Methods

To demonstrate the superiority of the proposed LiteLKG-Net, ten state-of-the-art RSCD methods were selected for comparison.

1) *Fully Convolutional–Early Fusion (FC-EF) [41]:* An end-to-end change detection model based on UNet architecture. The dual-temporal images are channelwise concatenated and fed into a single encoder–decoder for feature fusion and change map generation.
2) *Fully Convolutional–Siamese–Concatenation (FC-Siam-conc) [41]:* A Siamese CNN derived from FC-EF. It extracts bitemporal features independently and concatenates spatiotemporal features layerwise in the decoder during upsampling, ultimately generating pixel-level change maps.
3) *Fully Convolutional–Siamese–Difference (FC-Siam-diff) [41]:* It employs a Siamese encoder but computes absolute differences between feature maps at each decoding stage to directly output the change map.
4) *Bitemporal Image Transformer (BIT) [46]:* BIT integrates CNN and Transformer architectures, where the CNN extracts local features and the Transformer refines these features before forwarding them to a prediction head for final output.
5) *Spatial–Temporal Attention Network (STANet) [42]:* Uses Siamese encoders with basic attention module (BAM) and pyramid attention module (PAM) to capture global spatiotemporal dependencies for pixel-level change mapping.
6) *ChangeFormer [45]:* A Transformer-based Siamese network utilizing Transformer blocks, difference module, and MLP decoder for feature extraction and difference computation.
7) *Visual Change Transformer (VcT) [65]:* VcT integrates CNN with Transformer, enhancing feature representation through token mining and interactive modules to boost change detection accuracy.
8) *EATDer [66]:* A Transformer-based Siamese network comprising Siamese encoders and an edge-aware decoder with a joint loss function for high-quality change maps.
9) *Spatial–Temporal Robust Representation (STRobustNet) [67]:* Transforms inconsistent bitemporal features by generating spatiotemporal robust representations, improving efficiency while reducing feature confusion.
10) *Interactive and Supervised Dual-Mode Attention Network (ISDANet) [68]:* It integrates neighborhood aggregation and interactive attention modules, augmented by a supervised attention decoder, to achieve lightweight and high-accuracy change detection.

TABLE II
QUANTITATIVE RESULTS OF DIFFERENT CHANGE DETECTION METHODS ON WHU-CD DATASET (BEST VALUES ARE HIGHLIGHTED IN BOLD)

Method	OA(%)	F1(%)	IoU(%)	Pre(%)	Rec(%)
FC-EF*	97.61	69.37	53.11	71.63	67.25
FC-Siam-Diff*	95.63	58.81	41.66	47.33	77.66
FC-Siam-Conc*	97.04	66.63	49.95	60.88	73.58
BIT*☆	98.75	83.98	72.39	86.64	81.48
STANet*	98.71	83.50	71.67	85.60	81.49
ChangeFormer☆	99.28	90.45	82.56	91.34	89.58
VcT*☆	99.18	89.58	81.12	89.39	89.77
EATDer☆	98.58	90.01	81.82	91.32	88.74
STRobustNet*	99.32	90.89	83.29	92.92	88.94
ISDANet*	99.22	90.13	82.03	92.26	88.09
LiteLKG-Net*	**99.52**	**94.02**	**88.71**	**95.03**	**93.03**

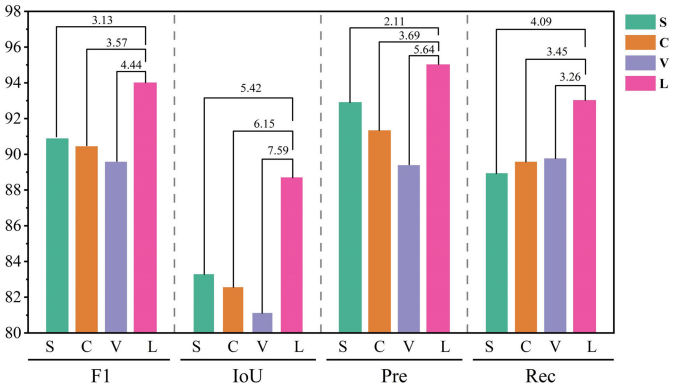


Fig. 9. Performance comparison on the WHU-CD dataset (S denotes STRobustNet, C denotes ChangeFormer, V denotes VcT, and L denotes LiteLKG-Net).

D. Results Evaluation

To validate the effectiveness and generalization capability of the proposed method, experiments were conducted on three public benchmark datasets: WHU-CD, LEVIR-CD, and SYSU-CD. Our approach was compared against ten state-of-the-art change detection methods, covering CNN-based (∗), Transformer-based (☆), and hybrid CNN–Transformer (∗☆) architectures. Both quantitative and qualitative analyses were conducted to evaluate model performance comprehensively. To ensure a fair and rigorous comparison and maintain experimental accuracy, all methods were trained and tested using identical datasets, iteration numbers, and other hyperparameters. The detailed experimental results are presented and discussed in Sections IV-D1–IV-D3.

1) Comparative Experiments on WHU-CD: The quantitative results of different methods on the WHU-CD dataset are shown in Table II. As observed, the proposed LiteLKG-Net outperforms all ten state-of-the-art comparative methods across OA, $F1$ score, IoU, precision, and recall metrics. Specifically, LiteLKG-Net achieves $F1$ score, IoU, and recall values of 94.02%, 88.71%, and 93.03%, respectively, exceeding the best competing method by 3.13%, 5.42%, and 4.26%. To further highlight the superiority of the proposed method, we selected three representative methods from different architectural categories: STRobustNet (denoted as S, CNN-based),

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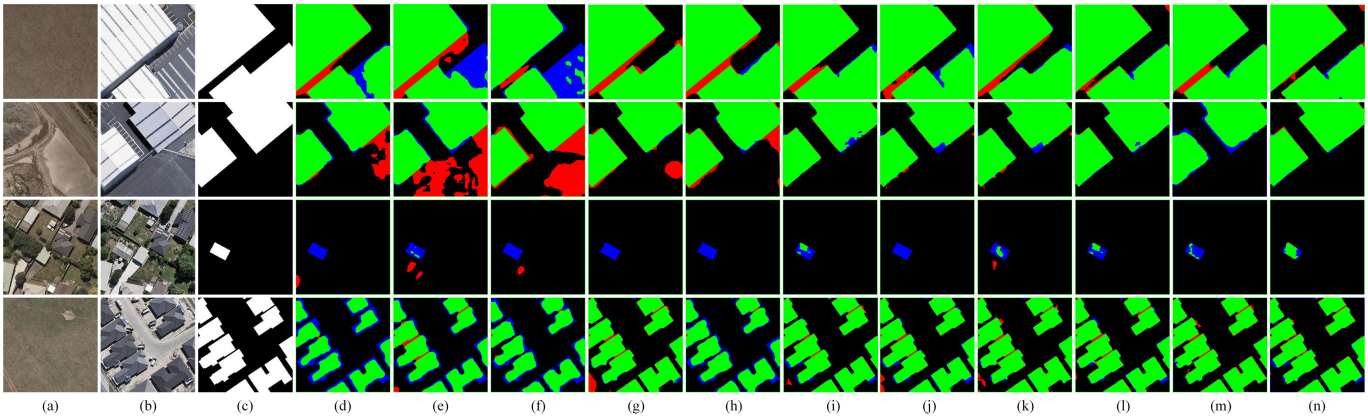


Fig. 10. Visual comparison results of different change detection methods on the WHU-CD dataset. (a) T1 images, (b) T2 images, (c) ground truth, (d) FC-EF, (e) FC-Siam-Diff, (f) FC-Siam-Conc, (g) BIT, (h) STANet, (i) ChangeFormer, (j) VcT, (k) EATDer, (l) STRobustNet, (m) ISDANet, and (n) LiteLKG-Net. In the change maps, green denotes TP, black denotes TN, red denotes FN, and blue denotes FP.

TABLE III
QUANTITATIVE RESULTS OF DIFFERENT CHANGE DETECTION METHODS ON LEVIR-CD DATASET (BEST VALUES ARE HIGHLIGHTED IN BOLD)

Method	OA(%)	F1(%)	IoU(%)	Pre(%)	Rec(%)
FC-EF*	98.39	83.40	71.53	86.91	80.17
FC-Siam-Diff*	98.67	86.31	75.92	89.53	83.31
FC-Siam-Conc*	98.49	83.69	71.96	91.99	76.77
BIT*☆	98.92	89.31	80.68	89.24	89.37
STANet*	98.66	87.26	77.40	83.81	**91.00**
ChangeFormer☆	99.04	90.40	82.48	92.05	88.80
VcT*☆	99.01	90.04	81.89	92.57	87.65
EATDer☆	98.75	91.20	83.83	91.74	90.67
STRobustNet*	99.10	91.11	83.66	91.54	90.67
ISDANet*	99.08	91.06	83.63	92.19	90.00
LiteLKG-Net*	**99.14**	**91.35**	**84.08**	**93.36**	89.42

ChangeFormer (C, Transformer-based), and VcT (V, hybrid architecture). As illustrated in Fig. 9, the proposed LiteLKG-Net (L) significantly outperforms these representative methods across four evaluation metrics. In particular, the $F1$ score, which best reflects the overall performance of the models, shows an improvement of up to 4.44% over the strongest competitors, demonstrating the effectiveness of the proposed approach.

To comprehensively evaluate the performance of all experimental methods, four representative scenarios from the test set were selected for comparative visualization, covering large, medium, and small change scales, as well as dense change scenes. Inference results on the WHU-CD test set are shown in Fig. 10. The proposed LiteLKG-Net outperforms ten comparative methods across multiple scenarios, with its results [see Fig. 10(n)] exhibiting the lowest error detection while preserving object integrity and edge details. Specifically, in small-scale dense buildings (see Fig. 10, row 4), FC-EF, FC-Siam-Conc, and FC-Siam-Diff, which adopt early or late fusion strategies and relatively simple structures, demonstrate the poorest performance, resulting in coarsest edge detection. STANet introduces a self-attention module during feature extraction, which yields higher edge accuracy than the former three models. The BIT model first utilizes a CNN backbone for feature extraction and then processes the features through a transformer structure. Compared to STANet, the multihead attention in the transformer block enhances the ability to capture building edge details. VcT, similar to BIT, employs a hybrid approach of CNN backbone and transformer attention mechanism. However, VcT further improves feature discrimination by mining representative background and change tokens from a coarse change map and introducing an anchor-guided attention mechanism for feature interaction and fusion. As a result, it achieves a 5.6% improvement in $F1$ score over BIT. ChangeFormer and EATDer, both based on pure transformer architectures, show significant improvements in detection accuracy compared to the aforementioned methods, with $F1$ scores exceeding 90%. Although STRobustNet incorporates the spatial–temporal robust (STR) representation module and the robust representation generation (RRG) module to enhance feature discriminability and stability, its performance still lags behind LiteLKG-Net. The MAM module in LiteLKG-Net focuses on capturing long-range dependencies and local detailed features, extracting multiscale contextual information hierarchically, and enhancing adaptability to complex scenes, thereby achieving superior change detection results. Finally, as shown in Fig. 10 (row 3), the compared models exhibit insensitivity to small-scale changes, which tends to result in missed detections of subtle changes.

2) Comparative Experiments on LEVIR-CD: Quantitative results of different methods on the LEVIR-CD dataset are presented in Table III. The proposed LiteLKG-Net outperforms ten state-of-the-art comparative methods in terms of OA, $F1$ score, IoU, and precision. Specifically, LiteLKG-Net achieves $F1$ score, IoU, and precision values of 91.35%, 84.08%, and 93.36%, respectively, surpassing the best-performing competitor by 0.15%, 0.25%, and 1.62% points. To further highlight the superiority of the proposed method, three representative approaches with distinct architectural paradigms were selected: STRobustNet (denoted as S, CNN-based), EATDer (denoted as E, Transformer-based), and VcT (denoted as V, hybrid architecture). As shown in Fig. 11, LiteLKG-Net (denoted as L) consistently outperforms all representative methods across the $F1$ score, IoU, and precision metrics. Notably, the $F1$ score—reflecting overall model

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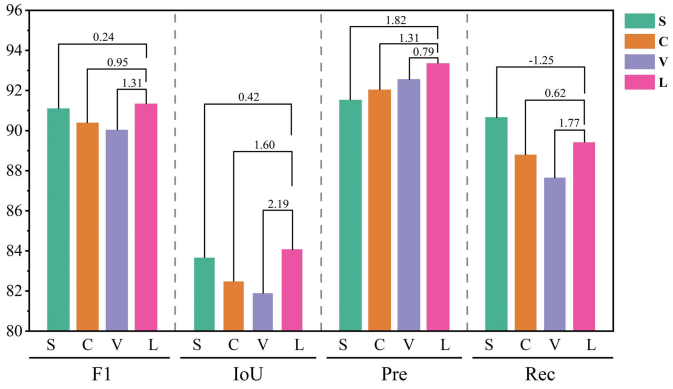


Fig. 11. Performance comparison on the LEVIR-CD dataset (S denotes STRobustNet, E denotes EATDer, V denotes VcT, and L denotes LiteLKG-Net).

TABLE IV
QUANTITATIVE RESULTS OF DIFFERENT CHANGE DETECTION METHODS ON SYSU-CD DATASET (BEST VALUES ARE HIGHLIGHTED IN BOLD)

Method	OA(%)	F1(%)	IoU(%)	Pre(%)	Rec(%)
FC-EF*	86.02	75.07	60.09	74.32	75.84
FC-Siam-Diff*	82.11	71.06	55.11	80.35	63.71
FC-Siam-Conc*	86.17	76.35	61.75	82.54	71.03
BIT*☆	90.18	78.15	64.13	74.49	82.18
STANet*	87.96	77.37	63.09	70.76	**85.33**
ChangeFormer☆	90.44	78.45	64.54	83.73	73.80
VcT*☆	90.08	79.08	65.40	78.67	79.50
EATDer☆	90.34	80.23	66.99	77.56	83.09
STRobustNet*	91.13	80.66	67.59	82.98	78.46
ISDANet*	91.68	82.21	69.72	80.52	83.97
LiteLKG-Net*	**91.93**	**82.27**	**69.89**	**85.47**	79.31

performance—achieves a maximum improvement of 1.31% points over the strongest competitors, thereby validating the effectiveness of the proposed approach.

The visual analysis of experimental results on the LEVIR-CD dataset is presented in Fig. 12. While EATDer, STRobustNet, and the proposed LiteLKG-Net demonstrate favorable performance on the test set, LiteLKG-Net produces predictions that are more consistent with the ground-truth labels. It achieves the highest prediction accuracy, with refined edges and superior performance across complex multiscale change scenarios. In large-scale structural change cases (see Fig. 12, row 1), comparative methods exhibit coarse boundaries and incomplete detection of change regions. In contrast, LiteLKG-Net leverages the MAM module in the encoder to aggregate global and local information, enabling the extraction of rich, multigranularity change features. The FFM further integrates multiscale features to enhance detail recognition. At the same time, the decoder employs a multilevel residual fusion strategy with an embedded GMM to adaptively select key features. This leads to significant improvements in detection accuracy and robustness.

For small-scale dense change scenarios (see Fig. 12, rows 2 and 3), LiteLKG-Net yields sharper boundary delineation, especially in distinguishing adjacent buildings. Finally, as illustrated in Fig. 12 (row 4), comparative methods demonstrate limited capability in detecting minute changes, often resulting in missed and false detections. These results, both quantitative and qualitative, affirm the superior performance of LiteLKG-Net over recently published ISDANet, STRobustNet, and EATDer.

3) Comparative Experiments on SYSU-CD: Quantitative results of different methods on the LEVIR-CD dataset are presented in Table IV. The proposed LiteLKG-Net outperforms ten state-of-the-art comparative methods in terms of OA, $F1$ score, IoU, and precision. Specifically, LiteLKG-Net achieves $F1$ score, IoU, and precision values of 82.27%, 69.89%, and 85.47%, respectively, surpassing the best-performing competitor by 0.06%, 0.17%, and 4.95% points. To further highlight the superiority of the proposed method, three representative approaches with distinct architectural paradigms were selected: ISDANet (denoted as I, CNN-based), EATDer (denoted as E, Transformer-based), and VcT (denoted as V, hybrid architecture). As shown in Fig. 13, LiteLKG-Net (denoted as L) consistently outperforms all representative methods across four evaluation metrics. Notably, the $F1$ score—reflecting overall model performance—achieves a maximum improvement of 3.19% points over the strongest competitors, thereby validating the effectiveness of the proposed approach.

The visual analysis of experimental results on the SYSU-CD dataset is presented in Fig. 14. The SYSU-CD dataset comprises complex change types, including newly constructed urban buildings, groundwork before construction, vegetation changes, road expansion, and offshore construction, which make the RSCD task particularly challenging. The visual comparisons between our method and other competing methods on the SYSU-CD dataset are shown in Fig. 14. The first row illustrates an offshore construction scenario, where it can be observed that the competing methods tend to produce more missed detections (false negatives) and exhibit certain false alarms (false positives) along object boundaries. In contrast, our method performs favorably in these aspects. By introducing the MAM and FFM, our method captures rich multiscale feature information and effectively fuses features at different scales, thereby enhancing the model's capability in recognizing change targets and optimizing boundary details. The second row shows a groundwork-before-construction scenario, in which the competing methods yield a large number of false alarms (false positives), while our method demonstrates superior performance.

The third row displays a vegetation change scenario. Due to the large and irregular nature of the changed areas, the competing methods suffer from extensive missed detections (false negatives). In contrast, our method accurately identifies the change targets and clearly delineates their boundaries, producing fine-grained predicted labels. The fourth row presents a road expansion scenario, where the changed objects are small and irregular in shape, posing a significant challenge to change detection. As shown in Fig. 14, the competing methods can only detect part of the changed areas, with many missed detections (false negatives) and some false alarms (false positives). In contrast, our method, despite exhibiting

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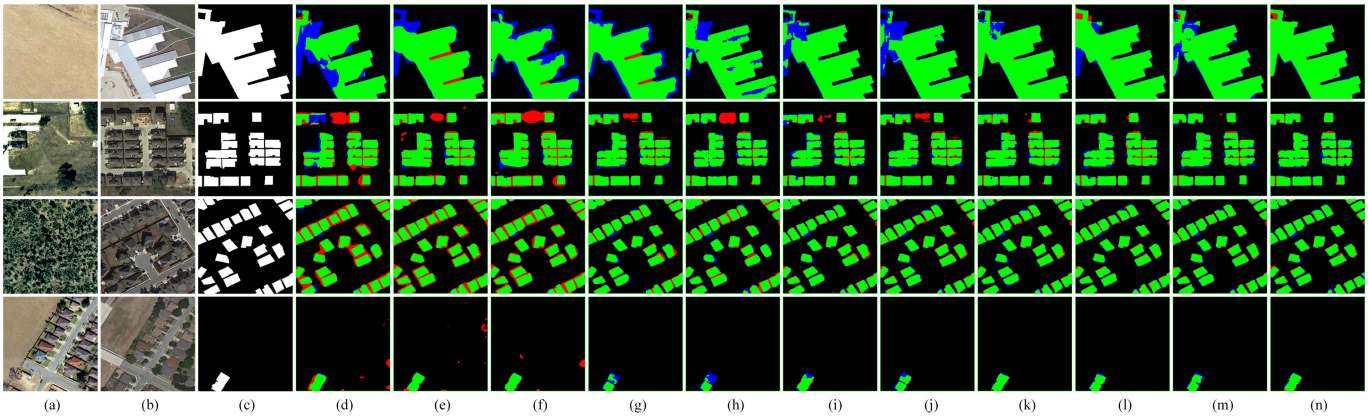


Fig. 12. Visual comparison results of different change detection methods on the LEVIR-CD dataset. (a) T1 images, (b) T2 images, (c) ground truth, (d) FC-EF, (e) FC-Siam-Diff, (f) FC-Siam-Conc, (g) BIT, (h) STANet, (i) ChangeFormer, (j) VcT, (k) EATDer, (l) STRobustNet, (m) ISDANet, and (n) LiteLKG-Net. In the change maps, green denotes TP, black denotes TN, red denotes FN, and blue denotes FP.

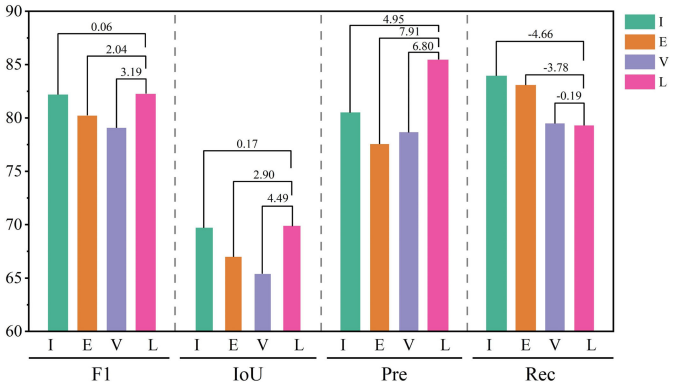


Fig. 13. Performance comparison on the SYSU-CD dataset (I denotes ISDANet, E denotes EATDer, V denotes VcT, and L denotes LiteLKG-Net).

TABLE V

PERFORMANCE COMPARISON OF LITELKG-NET (WITH/WITHOUT GAUSSIAN NOISE, $\sigma = 0.1$, APPLIED TO 30% OF IMAGES)

Dataset	Condition	F1(%)	IoU(%)
WHU-CD	Original (no noise)	94.02	88.71
	30% Gaussian Noise	93.39	87.59
LEVIR-CD	Original (no noise)	91.35	84.08
	30% Gaussian Noise	90.11	81.99
SYSU-CD	Original (no noise)	82.27	69.89
	30% Gaussian Noise	81.07	68.16

a small number of false alarms (false positives), achieves the best overall performance.

To further evaluate the robustness of LiteLKG-Net under complex scenarios, supplementary experiments were conducted on three datasets. For each dataset, we randomly selected 30% of the test image pairs, normalized them, and added Gaussian noise ($\sigma = 0.1$) to simulate potential sensor and environmental disturbances. The experimental results are summarized in Table V. On all three benchmarks, LiteLKG-Net exhibited only a slight performance degradation under noise corruption. Specifically, although Gaussian noise caused $F1$ score reductions of 0.63%, 1.24%, and 1.20% on the respective datasets, the overall performance remained stably high. This indicates that LiteLKG-Net possesses strong robustness and stability in complex scenarios.

TABLE VI

PERFORMANCE–COMPLEXITY TRADEOFFS OF DIFFERENT CHANGE DETECTION METHODS ON THE WHU-CD DATASET

Method	Params (M)	FLOPs (G)	AIT (ms)	FPS	F1(%)
FC-EF*	1.35	3.57	13.12	76.22	69.37
FC-Siam-Diff*	1.35	4.72	17.63	56.72	58.81
FC-Siam-Conc*	1.35	5.32	17.65	56.66	66.63
BIT*☆	3.49	10.63	32.67	30.61	83.98
STANet*	16.94	18.03	41.91	23.86	83.50
ChangeFormer☆	41.03	45.97	76.69	13.04	90.45
VcT*☆	3.57	10.64	47.28	21.15	89.58
EATDer☆	6.59	23.43	47.28	21.15	90.01
STRobustNet*	5.23	13.19	32.14	31.11	90.89
ISDANet*	6.93	6.76	21.01	47.59	90.13
LiteLKG-Net*	2.56	4.52	19.57	51.11	**94.02**

E. Model Computation Complexity

Beyond quantitative and qualitative analyses across datasets, we further evaluate the tradeoff between model complexity (in terms of parameters, FLOPs, average inference time (AIT), and FPS) and performance ($F1$ score) on the WHU-CD dataset to validate the efficacy of the proposed method. Table VI summarizes the parameters (Params), computational complexity (FLOPs), AIT, FPS, and $F1$ scores of all methods. The proposed LiteLKG-Net achieves the highest $F1$ score (94.02%) with only 2.56 M parameters, 4.52 GFLOPs, an inference time of 19.57 ms, and 51.11 FPS. As shown in Fig. 15, LiteLKG-Net incurs slightly higher computational cost than FC-EF and ISDANet (+ 0.95 and + 1.06 GFLOPs, respectively), while significantly outperforming them. Early-fusion methods (FC-EF, FC-Siam-Diff, and FC-Siam-Conc) benefit from structural simplicity, resulting in low complexity but also the lowest accuracy. Hybrid CNN–Transformer models (BIT and VcT) offer a reasonable tradeoff between

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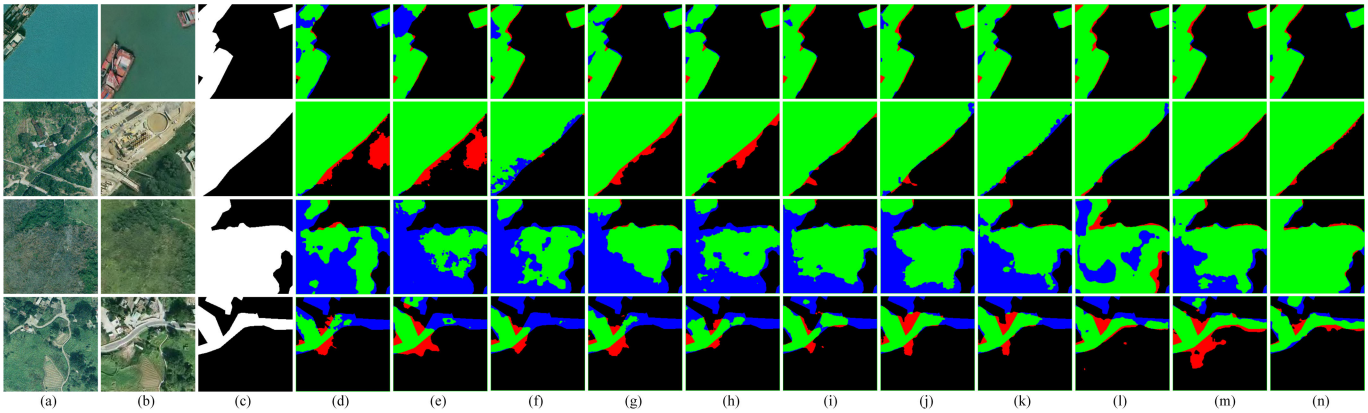


Fig. 14. Visual comparison results of different change detection methods on the SYSU-CD dataset. (a) T1 images, (b) T2 images, (c) ground truth, (d) FC-EF, (e) FC-Siam-Diff, (f) FC-Siam-Conc, (g) BIT, (h) STANet, (i) ChangeFormer, (j) VcT, (k) EATDer, (l) STRobustNet, (m) ISDANet, and (n) LiteLKG-Net. In the change maps, green denotes TP, black denotes TN, red denotes FN, and blue denotes FP.

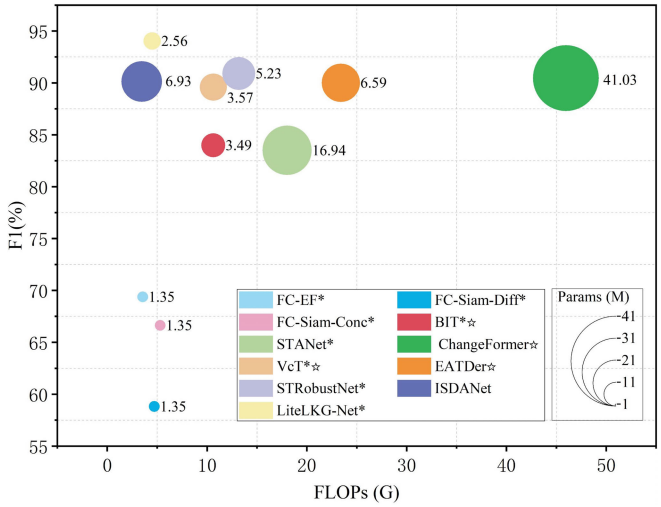


Fig. 15. Comparison of model complexity in terms of params, FLOPs, and $F1$ score for different methods on the WHU-CD dataset.

parameters and FLOPs, achieving moderate gains. In contrast, ChangeFormer and EATDer, being pure Transformer-based architectures, exhibit noticeably higher parameters and FLOPs compared to other methods. Although they achieve good performance, they fail to balance efficiency and effectiveness. LiteLKG-Net optimally balances complexity, inference efficiency, and accuracy—slightly exceeding FC-series models in parameters while significantly surpassing all competitors in performance—demonstrating best efficiency with minimal computational overhead.

In practical applications, the general workflow of change detection typically involves several steps: data acquisition, preprocessing (e.g., radiometric correction and coregistration), image tiling, model inference, and result mosaicking and visualization. Taking the WHU-CD dataset as an example, we analyzed the detection efficiency of the proposed method. Since the WHU-CD dataset has already been preprocessed by its authors (including coregistration and radiometric correction), our work primarily focuses on the model inference stage and its structural efficiency. On an NVIDIA RTX 2080Ti

TABLE VII
QUANTITATIVE PERFORMANCE COMPARISON OF ABLATION STUDY RESULTS BASED ON THE WHU-CD DATASET (BEST VALUES ARE HIGHLIGHTED IN BOLD)

Method	CIM	FFM	GMM	MAM	F1(%)	IoU(%)	Parms(M)	FLOPs(G)
Baseline					90.11	81.99	2.04	3.98
Baseline	✓				91.35	84.08	2.04	3.98
Baseline	✓	✓			93.41	87.64	2.32	4.14
Baseline	✓	✓	✓		93.52	87.82	2.56	4.45
Baseline	✓	✓	✓	✓	**94.02**	**88.71**	2.56	4.52

GPU, LiteLKG-Net achieves an AIT of 19.57 ms per 256×256 patch, and the complete inference for a 15k ×15k image (3481 patches) takes approximately 1 min. Considering the additional I/O overhead and tile mosaicking, the total inference time for a 15k ×15k bitemporal image is about 1 and 2 min. These results demonstrate that LiteLKG-Net can support city-scale (10k ×10k pixels) monitoring within several minutes, thereby meeting the efficiency requirements of real-world change detection applications.

F. Ablation Study

To comprehensively evaluate the effectiveness of each proposed module, ablation studies were conducted on the WHU-CD, LEVIR-CD, and SYSU-CD datasets. In these experiments, the proposed modules were incrementally added to the baseline model under the same experimental settings to assess their contributions. The results on different datasets are presented in Tables V–VII. The $F1$ score and IoU metrics validate the impact of each module on model performance and efficiency.

CIM: A lightweight CIM is introduced into the backbone network. CIM enables efficient feature interaction through periodic channelwise exchange between bitemporal feature maps, enhancing feature diversity and sensitivity to change while improving feature extraction capability. By fusing feature channels, CIM directs the network to focus on genuine changes between bitemporal remote sensing images, significantly enhancing robustness to pseudochanges, such

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TABLE VIII

QUANTITATIVE PERFORMANCE COMPARISON OF ABLATION STUDY RESULTS BASED ON THE LEVIR-CD DATASET (BEST VALUES ARE HIGHLIGHTED IN BOLD)

Method	CIM	FFM	GMM	MAM	F1(%)	IoU(%)	Parms(M)	FLOPs(G)
Baseline					89.94	81.72	2.04	3.98
Baseline	✓				90.21	82.50	2.04	3.98
Baseline	✓	✓			90.73	83.03	2.32	4.14
Baseline	✓	✓	✓		91.01	83.50	2.56	4.45
Baseline	✓	✓	✓	✓	**91.35**	**84.08**	2.56	4.52

TABLE IX

QUANTITATIVE PERFORMANCE COMPARISON OF ABLATION STUDY RESULTS BASED ON THE SYSU-CD DATASET (BEST VALUES ARE HIGHLIGHTED IN BOLD)

Method	CIM	FFM	GMM	MAM	F1(%)	IoU(%)	Parms(M)	FLOPs(G)
Baseline					78.36	66.21	2.04	3.98
Baseline	✓				80.31	67.10	2.04	3.98
Baseline	✓	✓			80.86	67.72	2.32	4.14
Baseline	✓	✓	✓		81.22	68.38	2.56	4.45
Baseline	✓	✓	✓	✓	**82.27**	**69.89**	2.56	4.52

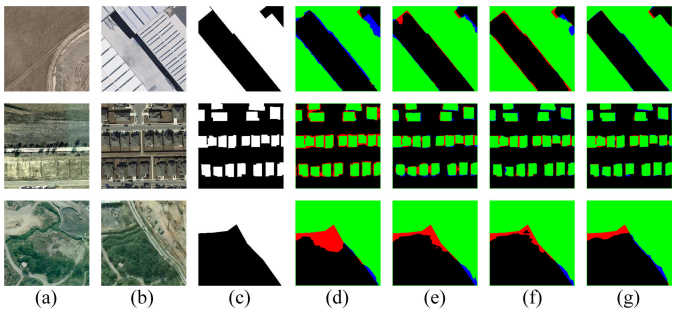


Fig. 16. (a) T1 phase image, (b) T2 phase image, (c) true label, (d) visualization result with CIM added, (e) visualization result with FFM added, (f) visualization result with GMM added, and (g) visualization result with MAM added. The samples are from WHU-CD, LEVIR-CD, and SYSU-CD datasets, respectively.

TABLE X

MODULE REPLACEMENT ABLATION RESULTS

Method	**F1(%)**	**IoU(%)**	**Params(M)**	**GFLOPs**	**FPS**
LiteLKG-Net (full)	94.02	88.71	2.56	4.52	51.11
MAM replaced by CBAM	92.80	86.57	2.56	4.47	51.96
FFM replaced by BiFPN	93.61	87.98	2.64	4.62	48.28

TABLE XI

MLKA HYPERPARAMETER SENSITIVITY ABLATION RESULTS

MLKA Config	**F1(%)**	**IoU(%)**	**Params (M)**	**GFLOPs**	**FPS**
small	93.26	87.37	2.552	4.479	51.26
medium	93.57	87.91	2.559	4.515	51.15
large	93.29	87.43	2.569	4.566	51.03
Progressive (Our)	**94.02**	**88.71**	2.560	4.520	51.11
aggressive	93.74	88.22	2.564	4.540	50.96

as illumination and seasonal variations. In addition, this channel-level reorganization introduces no extra parameters or convolutional operations, thereby preserving the model's lightweight nature while improving accuracy. As shown in Tables VII–IX, integrating CIM into the backbone improves the $F1$ score and IoU by 1.24%/2.09%, 0.27%/0.78%, and 1.95%/0.89% on the three datasets, respectively. As observed in Fig. 16(d), the model incorporating CIM accurately detects change targets but exhibits minor false positives and false negatives in edge details and localized regions.

FFM: An ablation study was conducted on the FFM to verify its effectiveness. FFM is designed to enhance efficient interaction between bitemporal features, thereby fully exploiting Semantic information from different stages. By employing a feature fusion strategy based on concatenation and residual learning, FFM enhances the model's ability to identify small-scale changes and boundary features, thereby improving change detection performance. Meanwhile, the number of parameters and FLOPs increases by only 0.28 M and 0.16 G, respectively, thereby maintaining the model's lightweight nature. As shown in Tables VII–IX, introducing FFM into the backbone after CIM increases the $F1$ score and IoU by 2.06%/3.56%, 0.52%/0.53%, and 0.55%/0.62% on the three datasets, respectively. As observed in Fig. 16(e), the model incorporating the FFM demonstrates refined edge detail processing, measurable reduction in false positives, and significant overall performance improvement, validating FFM's efficacy.

GMM: The GMM is introduced after the FFM and serves as a core component of the decoder architecture. GMM significantly enhances the discriminative capability of RSCD through a channel-adaptive feature calibration mechanism. It dynamically adjusts spatial feature weights using three learnable parameters (α,β, and γ), improving the model's responsiveness to change features while maintaining computational efficiency. Specifically, the channel importance scaling factor α is used to evaluate the significance of each channel, enabling effective utilization of channel information. The nonlinear bias term β emphasizes important channel features and suppresses less relevant ones. The parameter γ models the competition among channels, effectively reducing parameter redundancy. Finally, a gating function is employed to perform nonlinear adjustment and dynamic selection of channel weights, which significantly improves the model's ability to recognize boundaries and other complex change features. As shown in Tables VII–IX, incorporating GMM into the backbone after FFM improves the $F1$ score and IoU by 0.11%/0.18%, 0.28%/0.47%, and 0.36%/0.66% on the three datasets, respectively, with only 0.24 M and 0.31 G increases in parameters and FLOPs. As illustrated in Fig. 16(f), the introduction of GMM further enhances the model's ability to capture boundary details, resulting in finer segmentation of change edges, especially for dense small-scale change targets, confirming the effectiveness of GMM.

Multiscale Attentional Module (MAM): The MAM is a key component of LiteLKG-Net, designed to enhance change localization in remote sensing imagery by integrating multiscale contextual perception with dynamic spatial attention. MAM comprises two submodules: the MLKA and the GSAU. Specifically, MLKA employs multibranch

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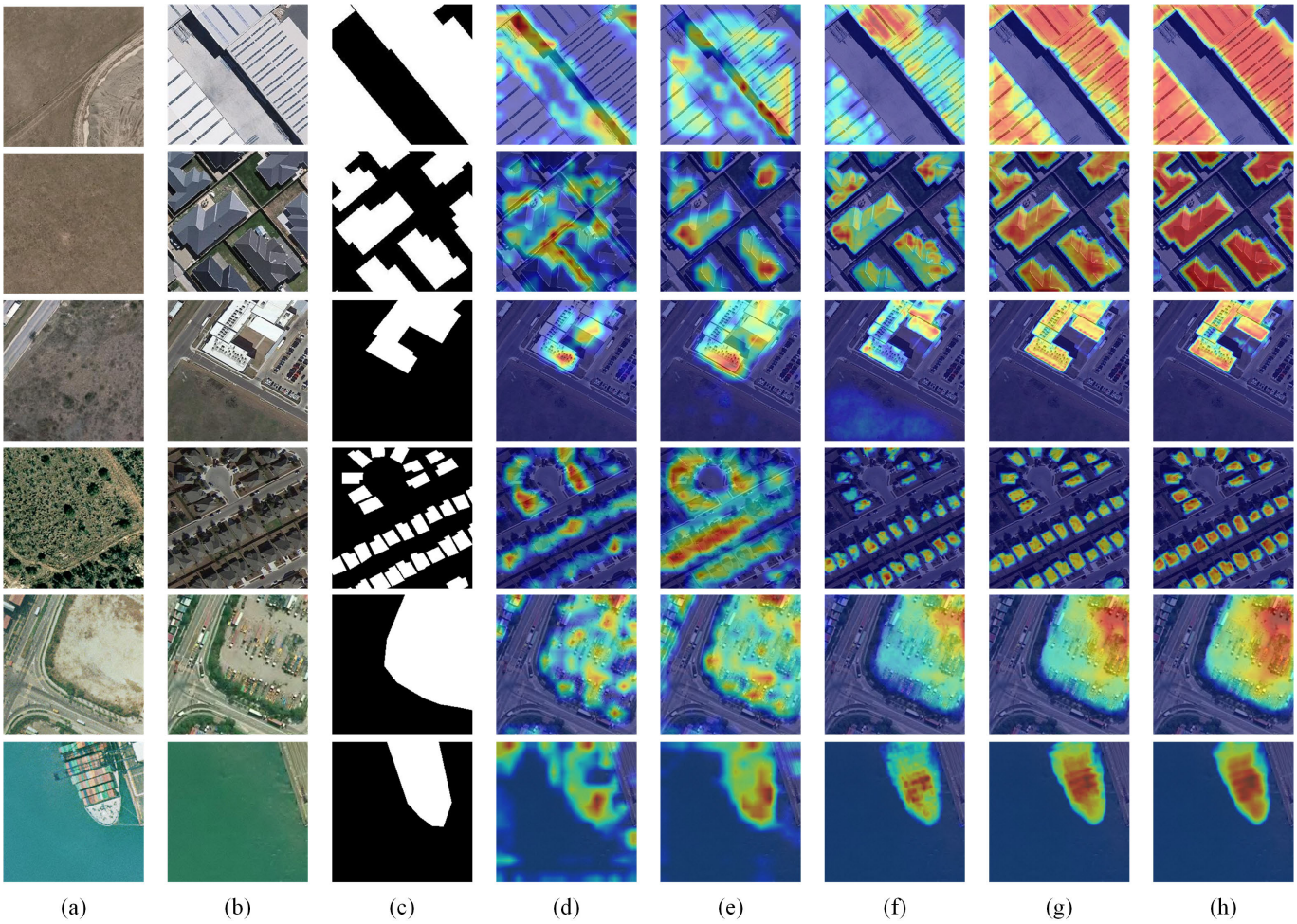


Fig. 17. Visualization of heatmaps at different stages. (a) T1 images, (b) T2 images, (c) ground truth, (d) visualization of the backbone, (e) visualization after adding CIM, (f) visualization after adding MAM, (g) visualization after adding FFM, and (h) visualization after adding GMM. Samples are taken from WHU-CD (rows 1 and 2), LEVIR-CD (rows 3 and 4), and SYSU-CD (rows 5 and 6) datasets.

dilated convolutions to capture both global dependencies and local details, improving the model's adaptability to complex scenes. GSAU leverages depthwise separable convolutions and a gating mechanism to generate dynamic spatial weights, suppressing noise and highlighting salient regions. Their synergy enables MAM to effectively balance computational cost and feature representation, significantly boosting detection accuracy and robustness within a lightweight framework. As reported in Tables VII–IX, integrating MAM after the FFM leads to $F1$/IoU improvements of 0.50%/0.89%, 0.34%/0.58%, and 1.05%/1.51% on the three datasets, with no additional parameters and only a 0.07 G increase in FLOPs. Visual results in Fig. 16(g) further confirm that MAM reduces false positives and false negatives, and enhances the delineation of change boundaries, validating its effectiveness.

Furthermore, we conducted module replacement and hyperparameter ablation experiments on the WHU-CD dataset. First, to verify the effectiveness of the attention mechanism and FFMs, we performed comparative ablation studies by replacing our proposed modules with well-established alternatives. Specifically, the MAM module was replaced with the advanced CBAM [69], and the FFM module was replaced with the BiFPN [70]. As shown in Table X, compared with CBAM, our MAM achieves a 1.22% improvement in $F1$ score, demonstrating the superiority of the multiscale large-kernel design in capturing complex spatial contextual dependencies for change detection. In addition, our FFM outperforms BiFPN by 0.41% in $F1$ score, while also exhibiting significant advantages in terms of model complexity and inference speed. We also conducted a comprehensive sensitivity analysis of the convolution kernel sizes in the three branches of MLKA. Five configurations were systematically evaluated:1) small: all small kernels, with each branch using (3, 5, 1); 2) medium: all medium kernels, with each branch using (5, 7, 1); 3) large: all large kernels, with each branch using (7, 9, 1); 4) progressive: the progressive configuration adopted in this article, (3, 5, 1), (5, 7, 1), and (7, 9, 1); and 5) aggressive: an aggressive configuration using (3, 7, 1), (5, 9, 1), and (7, 11, 1) for the three branches, respectively. As shown in Table XI, the progressive kernel configuration used in our MLKA module achieves the best overall performance while maintaining a good tradeoff between accuracy and computational complexity.

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G. Network Visualization

To better understand the architecture of the proposed LiteLKG-Net and visually illustrate feature information at each stage, we perform a stagewise visualization using bitemporal remote sensing image pairs from WHU-CD, LEVIR-CD, and SYSU-CD datasets (see Fig. 17).

Starting from the input images T1 and T2 [see Fig. 17(a) and (b)], the Siamese backbone [see Fig. 17(d)] mainly captures mid- and low-level features along with some high-level semantics. At this stage, the SR is low, resulting in detail loss, but the large receptive field enables the capture of wide-range contextual information, while the delineation of change boundaries remains imprecise.

Adding CIM [see Fig. 17(e)] further enhances the encoder's feature extraction capability by periodically exchanging channels of bitemporal feature maps, increasing feature diversity and sensitivity to change features. Consequently, the heatmap becomes more focused on change regions, background responses are reduced, and boundary recognition improves, though some noise remains.

With MAM [see Fig. 17(f)], the integration of multiscale contextual awareness and dynamic spatial attention enables precise localization of change regions. MLKA captures both global long-range dependencies and local details via multibranch dilated convolutions. At the same time, GSAU generates dynamic spatial weights using depthwise separable convolutions and a gating mechanism to suppress noise and focus on key areas. As a result, the model becomes more sensitive to small-object changes and fine boundaries, with clearer heatmap edges and improved detection of small changes.

Incorporating FFM [see Fig. 17(g)] further enhances the recognition of small-scale changes through cascaded and residual feature fusion, improving change detection performance while maintaining lightweight complexity. The heatmap shows clearer contours of change regions, shapes closer to the ground truth, and improved discrimination in small targets and complex textures.

Finally, adding GMM [see Fig. 17(h)] adaptively adjusts spatial feature weights via triple learnable parameters (α, β, and γ), enhancing the response to change features while maintaining computational efficiency. This results in more complete region filling, smoother and more continuous boundaries, lower false detections, and improved robustness.

V. Conclusion

In this article, we present a novel network for RSCD, termed LiteLKG-Net. Considering the characteristics of RSCD tasks, we introduce a CIM, an MAM, an FFM, and a GMM into a lightweight Siamese architecture to enhance detection accuracy, efficiency, and robustness. The proposed method effectively addresses data-level uncertainties (e.g., annotation errors and class imbalance) and model-level uncertainties (e.g., misclassification of pseudochanges, difficulty in detecting small-scale targets, and change boundaries). The Siamese architecture enables better utilization of rich bitemporal remote sensing image features. The CIM cyclically exchanges the channels of bitemporal feature maps, thereby enhancing feature diversity and the model's sensitivity to change features, improving feature extraction capability while maintaining model lightweightness. The MAM achieves precise localization of change regions in remote sensing imagery by integrating multiscale contextual perception with dynamic spatial attention optimization. The MLKA captures both global long-range dependencies and local fine-grained details through multibranch dilated convolutions, improving adaptability to complex scenes. The GSAU generates dynamic spatial weights using depthwise separable convolutions and a gating mechanism to suppress noise and focus on key regions. The synergy between MLKA and GSAU allows MAM to balance computational efficiency and feature representation capability within a lightweight framework, significantly improving change detection accuracy and robustness. The FFM fuses multistage feature information via cascade and residual learning strategies, enhancing the recognition of small-scale change features and edge details. It improves change detection performance with only an additional 0.28 M parameters and 0.16 G FLOPs, thus preserving lightweightness. The GMM dynamically adjusts spatial feature weights through triple learnable parameters, substantially improving the recognition of change boundaries and other complex features while maintaining computational efficiency. We conducted comprehensive experiments on three public benchmark datasets—WHU-CD, LEVIR-CD, and SYSU-CD. Experimental results show that LiteLKG-Net outperforms ten state-of-the-art change detection methods across five evaluation metrics. The $F1$ scores of LiteLKG-Net reach 94.02%, 91.35%, and 82.27% on WHU-CD, LEVIR-CD, and SYSU-CD, respectively. Moreover, visual results demonstrate that LiteLKG-Net has significant advantages in detecting object integrity, edge details, and small-scale targets.

In future work, we plan to conduct more in-depth research on both data and detected objects: 1) explore multisource data fusion, such as integrating optical and SAR data for change detection tasks; 2) extend current change detection to semantic change detection, distinguishing not only between changed and unchanged regions but also identifying the categories of change objects (e.g., buildings, roads, vegetation, and cropland); 3) investigate edge device deployment and systematic performance benchmarking; and 4) investigate the in-depth applications of weakly supervised, unsupervised, and long-time-series analysis in the field of change detection.

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